

OFES: Optimal Feature Evaluation and Selection for Multi-Class Classification

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Abstract

The complexity and accuracy of classification algorithms largely depend on the size and the quality of the feature set used to build classifiers. Feature evaluation and selection are critical steps to decide a small set of high-quality features to build accurate and efficient classifiers since low-quality features not only have negative impacts on classification results but also increase the complexity of classification algorithms. **Current popular feature selection algorithms are not sufficient in selecting a set of high-quality features and discarding low-quality features, especially for streaming data.** This paper proposes a novel and efficient approach, optimal feature evaluation and selection (OFES), to evaluate and select high-quality features for multi-class classification. OFES first measures the difference between any two classes based on the feature that is to be evaluated. Then, it defines two quantitative measures to evaluate quality of the feature and identify high-quality features. Applying OFES in a multi-class classification application that identifies users based on their arm movement patterns, we find when compared with other popular feature evaluation and selection approaches, such as Information Gain Feature Ranking and Random Projections with Matlab feature ranking, OFES identifies a set of high-quality features that improves the accuracy of classification regardless of different classification algorithms. It also demonstrates great scalability with the increase of

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number of classes and yields a higher accuracy of 95%.

Keywords: Feature Evaluation, Feature Selection, Classification

1. Introduction

Multi-class classification classifies more than two classes in which each instance of a class is mutually exclusive with other classes [1]. Ideally, in the design of the feature-based classification algorithm, we should only choose a small set of high-quality features to train a classifier. This is because a large set of features not only significantly increases the complexity of classification algorithms, but also reduces the accuracy of the classification as it may contain some low-quality features. Thus, to improve the accuracy of the classification, feature selection helps to identify high-quality features [2]. How to select a set of high-quality features remains a research challenge.

The existing popular feature selection algorithms such as Correlation Based Feature Subset Selection (CFSS), Information Gain Feature Ranking (IGFR), and Random Projection have some disadvantages and are not efficient enough. For example, IGFR fails to accurately differentiate attributes with large numbers of distinct values because of overfitting issues [3]; CFSS does not provide any comparison between the features [4]; Random Projection method assumes that data classes are normally distributed and lacks a mechanism to decide the number of features [5].

To optimize the feature selection, in this paper, we propose a novel method for feature evaluation and selection for multi-class classification. In feature-based classification, which is to label a data sample using a class label, the main goal of feature selection is to identify features that can maximally link the data sample to a true labels as well as maximally unlink the data sample to other labels. In other words, a high-quality feature should maximize the difference between the true label and other labels. In this paper, we propose a two-phase mechanism, including feature evaluation and feature selection, to identify high-quality features. In the feature evaluation phase, we first adapt a novel

measure, Normalized Root Mean Squared Difference (NRMSD) [6, 7] to evaluate the difference between classes based on a specific feature. Then, we define two measures, Farness Value (FV) and Farness Ratio (FR) to quantitatively
 30 evaluate the quality of a feature, i.e., maximizing the difference among all the classes [8, 9]. In the feature selection phase, we design three algorithms to identify a set of high-quality features based on FV and FR values. The small set of selected high-quality features reduces the complexity of classification, and
 35 also improves the accuracy of classification.

We apply our novel feature evaluation and selection mechanism in user identification, a typical multi-class classification application, which identifies users based on data collected from activity sensors deployed at users' wrist. We extract features based on the activity sensor data while they are walking. Then,
 40 we evaluate 96 commonly used features and identify 10 high-quality features by discarding about 90% of the total features. We also find that selecting the small portion of features will not have significant impact on the classification accuracy. Compared with several other popular feature evaluation and selection approaches in multi-class classification, including Information Gain Feature Ranking (IGFR) + **Correlation Based Feature Subset Selection (CFSS)** [4],
 45 Random Projections with Matlab feature ranking [10], our proposed mechanism selects a set of high quality features which achieves 95% accuracy while others achieve less accuracy and inconsistent results with the increase of number of class labels.

The contribution of the paper is three-fold. First, we proposed a novel
 50 feature evaluation and selection mechanism which consists of two phases and three measures for multi-class classification. Second, we applied the proposed mechanism in the user identification application based on the activity sensor data in order to evaluate our method. Third, we compared our approach with
 55 two other feature selection and evaluation algorithms, Information Gain Feature Ranking (IGFR) + **Correlation Based Feature Subset Selection (CFSS)** and Random Projections with Matlab feature ranking. OFES outperforms them in terms of accuracy and consistency.

The remainder of the paper is organized as follows. Section 2 overviews
60 our extensive literature review that covers the current state-of-the-art in feature evaluation and selection techniques for multi-class classification. Section 3 discusses our proposed feature evaluation and selection mechanism along with the rationale of our design. Section 4 illustrates an example of applying our approach on user identification application. Section 5 discusses the performance
65 results of our research along with the comparison of others. Finally, we conclude this paper in Section 6.

2. Background

Binary classification classifies data samples to one of the two labels, while multi-class classification is a classification with more than two classes [11].
70 Multi-class classification has many applications, such as text classification [12, 13], face recognition [14], image recognition [15], and biometric identification [16]. In multi-class classification problem, each sample can be assigned to only one class [1]. The goal is to separate the classes with respect to the specific features [17]. Multi-class classification is recognized as more challenging as the
75 classification accuracy may decrease with the increasing number of classes [18]. Many approaches [2, 19, 20, 21] are explored to improve the performance of multi-class classification algorithms. Among them, feature selection is recognized as one of the most important practices. Next, we present an overview of popular feature selection approaches for multi-class classification.

80 2.1. Feature Analysis in Multi-class Classification

Feature selection is an important preprocessing step in classification with high dimensionality of features in the dataset [2]. Good features are needed in order to separate the classes well. As the number of features increases, complexity of the classification also increases [22]. Hence for better classification results,
85 it is important to analyze the discriminability of these features as they directly affect the classification results [23, 24]. Some of the common feature selection and extraction approaches [25] are ReliefF [26], **Sparsity Based Feature**

Selection [27], Correlation Based Feature Subset Selection (CFSS) [28], Information Gain Feature Ranking (IGFR) [29], and Random Projections [5, 10].

90 Next, we give a brief introduction to each of them.

2.1.1. *ReliefF*

ReliefF is one of the popular feature selection methods that ranks the importance of features by weighing them based on their quality or relevance [26]. The feature relevance is determined by how well data instances are separated. For
95 each data instance, Relief algorithm finds the nearest data point from the same class and nearest data points from different classes. This method is considered to be efficient because of low computational complexity. However, one of the drawbacks of this method is that it is sensitive to noise.

2.1.2. *Sparsity Based Feature Selection*

100 **Sparsity Based Feature Selection** is used to filter irrelevant or redundant features and reduce data dimensionality [30, 31, 32]. For example, Yu et al. [33] have used group sparsity method [34] to overcome the issue of more than enough features for classification. This method has the following advantages: Sparsity inducing algorithms are resistant to data noise [35], feature correlation can be taken advantage of
105 by using sparsity-based feature selection [36], and a sparsity-inducing regularization term is applied to the fitting error [32]. However, since most existing Structured Sparsity-inducing Feature Selection (SSFS) methods have higher time complexity of dimensional (d), scaling up
110 with high dimensionality is time-consuming and difficult [35].

Sparsity Based Feature Selection methods are mostly used to select features in image and biology datasets [37]. The optimal feature selection problem is NP-hard. Therefore, researchers adapted the problem to be a relaxed sparsity optimization problem [38]. Approximate and heuristic algorithms are proposed to find the solutions
115 of selecting a set of high-quality features. In our paper, we used a

heuristic algorithm to select high-quality features. For classification experiments, we used the Support Vector Machine model which has a reputation for producing sparse solutions [31]. In other words SVMs generalize better with a small set of data. Our approach is used in streaming continuous datasets. The goal of the sparsity based methods is to reduce the redundancy in the dataset [37]. Therefore, this approach could be complementary to our proposed approach because we select high-quality features from different perspectives.

We observed the following differences when comparing our approach with one of the sparsity based feature selection methods, namely, Sparsity Score. The Sparsity Score (SS) is a metric that assesses features based on their capacity to preserve sparsity [27]. Liu et al. [27] built a $l1$ graph using sparse representation on samples, and a $l1$ -norm based optimization is employed to concurrently construct the graph adjacency structure and related graph weights. SS method can be applied to both supervised and unsupervised, whereas our approach is supervised. Number of classes and sample size in their experiments is very less in comparison to our experiments. Time complexity of both the approaches are equal to $\mathcal{O}(n^2)$. The experiments they conducted are with data that is not continuous and use linear classification, whereas our method is designed for activity data or streaming sensory data applications. Hence, we can use this method as complementary once we normalize the data using sliding window technique. In future, we will review the results of our approach when combined with sparsity based feature selection. We will also modify our approach and make it feasible with more types of datasets, and compare the performance of our approach with sparsity based approaches using benchmark datasets.

145 *2.1.3. Correlation Based Feature Subset Selection (CFSS)*

CFSS is a method that ranks subsets of features by a correlation based heuristic evaluation function [28]. A feature is considered as a good one if it is relevant to the target concept, however not redundant to any other relevant features. The measure of goodness is determined by a correlation between features. 150 CFSS selects the subset of features which have the highest measure. Hence, the chosen subset of features have high correlation with the class and unrelated to each other [25]. Kumar et al. [4] applied CFSS method implemented in WEKA, with five search methods such as the best first (BF), genetic search (GS), greedy stepwise (GRS), linear forward selection (LFS), and subset size forward selection 155 (SSFS). These search methods use somewhat different mechanisms to choose the resulting subset. The authors also showed the correlation among the subset of features from the set of acceleration and rotation based features by using the CFSS method. However, the CFSS method fails to provide any comparison between the features with respect to the classes. On the contrary, our design 160 of feature evaluation shows the evaluation of each feature with respect to the classes.

2.1.4. Information Gain Feature Ranking (IGFR)

Another popular feature selection technique is to calculate the information gain. In this method, the information gain (also called entropy) for each attribute for the output variable is calculated. The values range from 0 (no information) to 1 (maximum information). The attributes that contribute more information will have a higher information gain value and can be selected, whereas the attributes that do not add much information will have a lower value and can be removed. In [4], along with CFSS method, Kumar et al. used IGFR 170 method to compensate for the drawback of CFSS by providing a measure of discriminability for each feature separately and ranking them accordingly with respect to the class label. They relied on attribute evaluator InfoGainAttributeEval, which is a WEKA implementation for information gain-based feature selection that measures how each feature contributes in decreasing the overall

175 entropy. Hence, this method evaluates the rank of the feature by computing the information gain of each feature with respect to the class. Conversely, this method does not work well for attributes with large number of distinct values because of overfitting issue, and it fails to accurately discriminate among the attributes [3].

180 2.1.5. *Random Projection*

Random projection is a simple technique used to reduce the dimensionality of a set of points that lie in Euclidean space. It is well-known for its power and low error rates when compared to other methods. Damasevicius et al. [5, 10] proposed a method for human activity recognition and user identification using 185 random projections to reduce the dimension of features so that they improve the efficiency of classification by lowering dimensionality feature space [5, 10]. In their proposed model, the best random projection with the smallest overlapping area is selected. Then, they selected the top three [5] or top ten [10] best features ranked by the Matlab Rankfeatures function using the entropy function. 190 Rankfeatures in Matlab ranks key features by class separability criteria. The good thing about this function is features that are highly correlated with already selected features are less likely to be included. However, the disadvantage is that it assumes that data classes are normally distributed which is not always the case. Therefore, their feature evaluation approach is not quite stable and 200 efficient, and there also lacks a mechanism to decide the number of features, which results in missing of many high-quality features.

From the analysis, we find that the above mentioned feature selection algorithms are mostly common used algorithms used in feature evaluation and selection for classification. These methods have some disadvantages and are not 200 efficient enough. Some of the problems of these techniques are lack of method for the comparison between the features, and overfitting issue. Our design of feature evaluation shows the evaluation of each feature with respect to the classes. Our method provides a mechanism to decide the number of features. Our fundamental goal is to find a minimum set of high-quality features by eliminating

205 low-quality features based on the feature evaluation results.

3. The proposed optimal feature evaluation and selection algorithm

The purpose of this research is to propose an optimal solution for feature evaluation and selection for classification. The quality and quantity of features selected affects the accuracy and efficiency of classification respectively. Hence, it is important to evaluate the features efficiently and select high-quality features. In the rest of this section, we first present the rationale of feature evaluation. Then, we define new measures to evaluate the features efficiently. Finally, we introduce a mechanism to select the set of high-quality features based on the evaluation.

215 3.1. Rationale of Feature Evaluation

Identifying features from the set of extracted features that can increase the separation between the classes is one of the most important preprocessing steps for classification [39]. On one hand, the quality of selected features mainly decides the accuracy of classification results. In order to achieve highly accurate classification, we need high-quality features that can distinctly differentiate any two classes. The difference between one class and other classes is not significant when they are compared using a poor feature. On the other hand, the number of features impacts the efficiency of the classification algorithm. Usually, a single feature may not be sufficient to classify a class label. Most previous studies use a vector of features in their algorithms [40, 41], but increasing the number of features increases both the dimension and the size of the dataset, which complicates the classification algorithm. Thus, we aim to find a minimum set of high-quality features by eliminating poor features based on the feature evaluation results.

230 When different features are used to build classifiers, they will have different results in classification accuracy as they may exhibit various capabilities in differentiating two different classes. We elaborate our idea using observations

from an example of 14-class classification, whose goal is to differentiate each of the 14 class labels from others accurately [9]. Let us consider 2 features, Mean and Standard Deviation on the z-axis data of the gyroscope. For each feature, we can measure and visualize the distance/difference between any two classes represented by two sets of data. For example, we can use a 14 X 14 matrix to display the distance/difference of any two classes in regards to one specific feature.

Figure 1 illustrates an example of visualized distance matrix of 14 class labels for these 2 features. In the figures, each square represents the measure of distance between a pair of classes for the evaluating feature. The diagonal squares represent the value of distance calculated based on two sets of data for the same class and other squares represent the value of distance calculated based on two sets of data from two different classes. For example, square (3,3) depicts the distance between two sets of data collected from class 3, and square (4,3) depicts the distance between datasets of class 4 and class 3. We use a darker color to represent a smaller distance value, which indicates a high level of similarity, while the lighter color represents a relatively bigger distance value, which indicates a low level of similarity. Therefore, a high-quality feature will result in a darker color for diagonal squares in comparison to a lighter color for other squares in the figure.

Examining the two sub-figures, Figure 1(a) and Figure 1(b), we have two observations. First, if the color of other squares in the figure is very similar to the diagonal squares, the feature may not be good enough to differentiate classes. Therefore, Standard Deviation(b) is a better feature than Mean(a). Second, a single feature is usually not sufficient to distinguish other classes from one specific class, because other classes may exhibit a high level of similarity on that feature. For example, observing from the figure, although Standard Deviation is a good feature, class 1 and class 9 exhibit a high level of similarity on Standard Deviation. For a better classification result, we need to consider multiple features together to distinctly differentiate classes.

The results in Figure 1 are obtained from analysis of a set of activ-

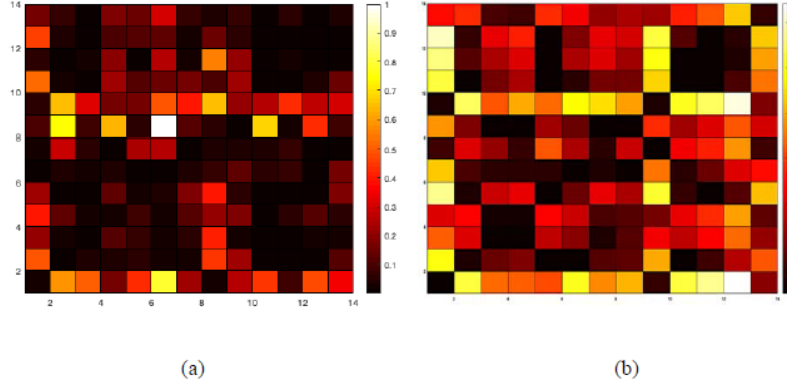


Figure 1: Visualization of class distance for single feature (a)Mean on gz-axis (b)Standard Deviation on gz-axis.

ity data collected from 14 different users. The results do not conclude
 265 **SD is always a better feature than Mean. Instead, the results motivate us to conduct this research, i.e., based on data collected from a specific application, some features may be better than others in differentiating two classes. In this example, we observe that based on collected activity data, SD is a better feature than Mean in identifying different users. This only applies to this specific application based on the specific type of data. When we choose a different dataset from other applications, the conclusion may vary.**

Figure 2 shows the results of two distance matrix when features are aggregated together, the combination of Kurtosis and Skewness (Figure 2(a)) and
 275 the combination of Mean and Standard Deviation (Figure 2(b)). **This can be achieved by just adding the two NRMSD matrices, i.e., Kurtosis and Skewness, which results in a combined NRMSD matrix.** From both figures, Figure 1 and Figure 2, we have following observations. First, even though Mean alone is not a better feature compared to Standard Deviation in Figure 1,
 280 when they are aggregated together, they make a better feature in Figure 2(b). Second, some features might not be good when combined with other features, for instance the combination of Kurtosis and Skewness in Figure 2(a).

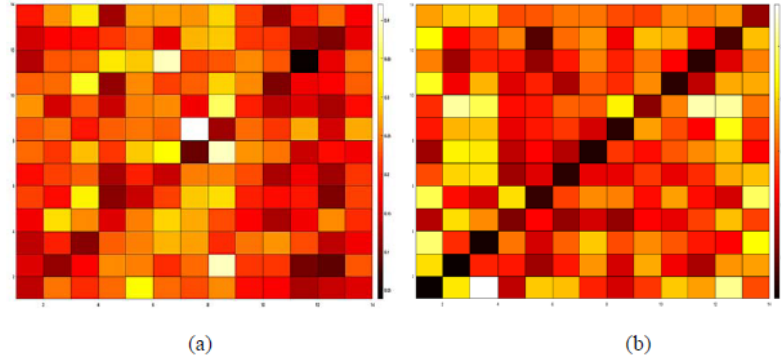


Figure 2: Visualization of class distance for combined features (a)Kurtosis+Skewness (b)Mean+Standard Deviation.

3.1.1. Summary of observations from distance matrix

Based on the above analysis, we conclude our observation on the relationship
 285 between a high-quality feature and the difference matrix as follows. These
 observations guide our design of feature evaluation and selection mechanisms
 to identify a set of high-quality features.

Observation 1. A high-quality feature identifies similarity in datasets of the
 same class and shows significant differences existing in different classes like that
 290 in Figure 1(b).

Observation 2. A low-quality feature is harder to differentiate a class from other
 classes as indicated by the color of other elements close to or darker than that
 of the diagonal elements in Figure 1(a).

Observation 3. Aggregating two or more features may yield better results in
 295 most cases as shown in Figure 2(b).

Observation 4. If a feature is of low-quality, aggregating it with other features
 may not yield better classification results as indicated in Figure 2(a).

Hence, by measuring the distance between the classes for a particular feature,
 300 we can clearly identify the high-quality features. Next, we define a measure
 named Normalized Root Mean Squared Difference (NRMSD) to measure the
 distance between the classes for a specific feature.

3.2. Class distance measured by Normalized Root Mean Squared Difference (NRMSD)

We adapt Normalized Root Mean Squared Difference (NRMSD) [6, 7] to
 305 measure the distance between two sets of data with respect to the features.
 The NRMSD is calculated based on two sets (set a and set b) of feature data.
 NRMSD is defined as in Equation 1, where a_i is the i^{th} data point in set a and
 b_i is the i^{th} data point in set b, and w is the size of the dataset.

$$NRMSD = \frac{\sqrt{\sum_{i=1}^w (a_i - b_i)^2}}{\sqrt{\sum_{i=1}^w (a_i)^2} + \sqrt{\sum_{i=1}^w (b_i)^2}} \quad (1)$$

NRMSD is similar to Euclidean distance but is normalized by magnitude.
 310 The value of NRMSD ranges from zero to one. **If the two datasets are ex-
 exactly the same, then the NRMSD value will be zero. Otherwise, the
 NRMSD value increases with the difference between two datasets.
 We can use this to differentiate classes, because given two sets of
 data, for high-quality features, if the datasets come from two differ-
 315 ent classes, the distance represented using NRMSD will be larger
 when compared with two sets of data coming from the same class.**
 Calculating the NRMSD for every pair of datasets of classes, we can build a 2-
 dimension $n \times n$ NRMSD matrix for n number of classes. The values of diagonal
 elements are expected to be zero or low for high-quality features. Non-diagonal
 320 elements i, j , (where $i \neq j$) in the matrix represent the NRMSD values of class i
 and class j. The values of these elements are expected to be high for high-quality
 features.

Similarly, the diagonal element i, j, (where $i=j$) in the matrix represents the
 NRMSD of two different datasets from the same class. For example, square
 325 (1,1) represents the NRMSD of two datasets for class 1. The NRMSD value will

not be zero as the two datasets will not be exactly the same although they are from the same class.

From above, there are two cases in calculating NRMSD in the NRMSD matrix. First, when calculating NRMSD for a specific class, we use two different datasets of the same class so that feature evaluation constitutes relatively more information. Second, when calculating NRMSD for two different classes, we use one dataset from each class. In our experiments, to differentiate classes, we use two datasets captured at two different times for all classes while calculating NRMSD. In other words, datasets of set a and b in Equation 1 are captured at different times. Hence, in this case, the non-diagonal element i, j (where $i \neq j$) is not the same as the non-diagonal element j, i . For example, as mentioned in Section 3.1, square (4,3) depicts the distance between datasets of class 4 and class 3. Since for each class we use two different datasets in our experiments, square (4,3) is not the same as square (3,4) in the NRMSD matrix. Similarly, the diagonal element i, j , (where $i = j$) in the matrix represents the NRMSD of two different datasets from the same class. For example, square (1, 1) represents the NRMSD of two datasets for class 1. The NRMSD value will not be zero as the two datasets will not be exactly the same although they are from the same class.

Since NRMSD is normalized by the magnitude of the features, it is very sensitive to changes in amplitude and phase. Although a small part of the data may be enough to evaluate a feature, it is important to have at least a full cycle of data recorded. Thus, two factors are important to consider when we use NRMSD to evaluate the feature, including the alignment of the data points in two datasets and the number of data points in the data set used in NRMSD calculation. The first requires to find two sets of data representing the two classes are aligned, and the second requires that the data set contains sufficient information representing the class. We can use Algorithm 1 to compute NRMSD and construct NRMSD matrix accordingly for a feature.

In algorithm 1, we calculate the distance between any two classes for the specific feature. To do so, we apply a sliding window approach here. **Hence, our model is applicable only for the streaming continuous data applications in the current status.** First, we set the window size. **The window size and sliding interval in the sliding window algorithm depends on various parameters, such as the number of samples in a dataset, the cycle of the dataset, etc. The process of finding these optimal values for window size and sliding interval is described in [42].** Then, for every window of feature data of class a and b , we calculate Euclidean distance. Similar to that in phase one, we slide windows in the two feature sets by incrementing one element so that for each pair of class (a,b) , we are able to find two windows with a best match, i.e., two windows that have the smallest Euclidean distance, which means that they are aligned and class a and b are closer. Next, we simply compute the NRMSD value for the best matched subset based on Equation 1. Continuing with this process, we can finally construct NRMSD Matrix with the NRMSD values of all pair of classes. **Time complexity of the ConNRMSD algorithm is $\mathcal{O}(k^2 * n^2)$, where k is the total number of classes and n is the total number of windows. If $k \ll n$, time complexity is equal to $\mathcal{O}(n^2)$.**

Our approach is not proposed to solve the class imbalance problem; however, based on preliminary analysis, our approach can work well with imbalanced datasets. First, because of using the sliding window approach, the data samples can be reused in different windows. Second, one property of high-quality features is that these features only need a small set of data to differentiate two different classes. As a result, the minimum class may provide sufficient samples that can be used in the training set. In addition, existing approaches such as [43, 44] can help to prepare balanced training sets. Hence, they are complementary to our proposed approach.

Algorithm 1 ConNRMSD

```
1: procedure CALCULATENRMSDMATRIX
2:   for a=1 to n do
3:     for b=1 to n do
4:       Step 1: Compute Euclidean distance for any two windows or
5:       subsets of feature data set each from class  $a$  and class  $b$ 
6:       Step 2: Find the best matched subsets with minimum Euclidean
7:       distance
8:       Step 3: Compute the NRMSD for class  $a$  and class  $b$  using
9:       equation 1
10:    end for
11:  end for
12:  Step 4: Construct  $NRMSDMatrix$  with the NRMSD values of all pair
13:  of classes
14:  return  $NRMSDMatrix$ 
15: end procedure
```

3.3. Definition of measures for feature evaluation

The NRMSD matrix mentioned in the above section, provides us an intuitive way of finding high-quality features through visualization like what we have presented in Figure 1 and 2. However, we still need to define measures that can quantitatively determine if a feature is a high-quality feature. Here, we define two heuristics, Farness Value and Farness Ratio, as the measures to evaluate features based on NRMSD matrix.

3.3.1. Farness Value (FV)

The first measure, Farness Value measures the average of the differences between other elements (other classes) and the diagonal elements (same class) in the NRMSD matrix. The formula to calculate the Farness Value is given below.

$$FV = \frac{\sum_{u=1}^n \sum_{v=1}^n C_{uv} - C_{uu}}{n(n-1)} \quad (2)$$

where,

- u : a candidate class (fixed)
- v : other classes (variable)
- n : number of classes
- C_{uv} : NRMSD value of class u and class v
- C_{uu} : NRMSD value of the same class

Time complexity of the Farness Value algorithm is $\mathcal{O}(k^2 * n^2 + k^2)$, where k is the total number of classes and n is the total number of windows. If $k \ll n$, time complexity is equal to $\mathcal{O}(n^2)$.

The Farness Value is a good heuristic because it considers both the average NRMSD value for the pair of same class and the average NRMSD value of the pair of different classes. Algorithm 2 gives details on how to calculate the Farness Value in five steps. Firstly, the two-dimensional NRMSD matrix is computed based on Algorithm 1. Then, we make each element of a column subtract the diagonal element at the same column in Step 2, 3 and 4. Finally, we calculate the Farness Value as the average of the non-diagonal elements. It ranges from

0 to 1. The bigger the Farness Value, the higher the quality of the feature,
 415 because a bigger Farness Value indicates a significant difference between a class
 and other classes.

Algorithm 2 Farness Value

```

1: procedure CALCULATEFARNESVALUE
2:   Step 1: Compute the two-dimensional ( $n \times n$ ) NRMSD matrix as matrix
3:   1 using Algorithm 1
4:   Step 2: Define a two-dimensional ( $n \times n$ ) matrix 2 identical to matrix 1.
5:   Step 3: Update matrix 2 by setting the value of elements at each column
6:   to the value of the diagonal element at that column
7:   Step 4: Subtract matrix 2 from matrix 1
8:   Step 5: Sum all the elements in the updated matrix from step 4 and
9:   divide by  $n \times (n-1)$  and store as FarnessValue
10:  return FarnessValue
11: end procedure

```

3.3.2. Farness Ratio (FR)

The second measure, Farness Ratio is defined as the average of the geometric
 average of column element to diagonal element ratio and the geometric average
 420 of row element to diagonal element ratio. The formula to calculate Farness
 Ratio is given below.

$$FR = \frac{\sqrt[n-1]{\prod_{v=1}^n \frac{C_{uv, u \neq v, 1 \leq u \leq n}}{C_{uu}}} + \sqrt[n-1]{\prod_{v=1}^n \frac{C_{vu, u \neq v, 1 \leq u \leq n}}{C_{uu}}}}{2} \quad (3)$$

where,

- u : a candidate class (fixed)
- v : other classes (variable)
- 425 - n : number of classes
- C_{uv} : NRMSD value of class u and class v
- C_{uu} : NRMSD value of the same class

Time complexity of the Farness Ratio algorithm is $\mathcal{O}(k^2 * n^2 + k^2)$,

where k is the total number of classes and n is the total number of
430 windows. If $k \ll n$, time complexity is equal to $\mathcal{O}(n^2)$. In Equation 1,
the NRMSD value is calculated based on two sets of streaming data.
 C_{uu} is calculated based on two sets of data collected from the same
class (i.e., two sets of activity data collected from the same person at
different times). C_{uu} equals zero only when the two datasets are fully
435 overlapped. The probability is close to zero. Therefore, C_{uu} will not
be zero.

Algorithm 3 presents the detailed steps to calculate Farness Ratio. The
first step is similar to the first step of Algorithm 2. In Step 2 and 3, for each
column, we divide each cell with diagonal element of that column in the NRMSD
440 matrix and calculate the product of the cells of each column. Then, we calculate
 $(n-1)^{th}$ root of the product for each column. In Step 4, we calculate the average
of these values of all the columns. Similarly, in Step 5 and 6, for each row, we
divide each cell with diagonal element of that row in the NRMSD matrix and
calculate the product of the cells of each row and calculate $(n-1)^{th}$ root of
445 the product for each row. In Step 7, we calculate the average of these values
of all the rows. Finally in Step 8, we calculate the Farness Ratio by taking the
average of values in Step 4 and 7. Likewise, the bigger the Farness Ratio, the
higher the quality of the feature.

3.4. Feature Selection

450 Extracting high-quality features from the raw dataset not only reduces the
dimension of the data, but also helps prepare a set of more meaningful data.
Based on the results of our feature evaluation method, we select a set of high-
quality features for classification. We use the following three approaches for
Feature Selection.

455 3.4.1. Based on the special values of Farness Value and Farness Ratio

First, we observe the range of Farness Value of all the extracted features.
Then, we set the threshold to FV and analyze which features meet the criteria.

Algorithm 3 Farness Ratio

```
1: procedure CALCULATEFARNESSRATIO
2:   Step 1: Compute the two-dimensional ( $n \times n$ ) matrix using Algorithm 1
3:   Step 2: For each cell in a column ( $C_{uv}$ ), divide by ( $C_{uu}$ )
4:   Step 3: Take  $(n - 1)^{th}$  root of the product of all cells in a column ( $C_{uv}$ )
5:   Step 4: Take the average of all the values from Step 3
6:   Step 5: For each cell in a row ( $C_{vu}$ ), divide by ( $C_{uu}$ )
7:   Step 6: Take  $(n - 1)^{th}$  root of the product of all cells in a row ( $C_{vu}$ )
8:   Step 7: Take the average of all the values from Step 6
9:   Step 8: Take the average of the values from Step 4 and Step 7 as
10:  FarnessRatio
11:  return FarnessRatio
12: end procedure
```

Second, we follow the same procedure for Farness Ratio and set the threshold to FR. Finally, we consider both FV and FR together, and select features that meet the threshold for both FV and FR as high quality features because the bigger FV and FR value implies significant difference between two classes. Our goal here is to attain a smaller set of high-quality features. Hence, when we consider both measures together, we will achieve more efficient results.

3.4.2. Based on ranking of Farness Value and Farness Ratio

For this approach, first, we rank FV of all the extracted features from the highest to the lowest. Similarly, we rank FR for all the extracted features. For each feature, we calculate the average of the ranks of FV and FR for that feature. Then, we rank the value of averages and arrange the ranks in ascending order. We select the required number of features from the top ranked features list considering the complexity of classification, number of classes, size of data set and resource limitation.

3.4.3. Based on both special values and ranking of Farness Value and Farness Ratio

For this approach, we first apply both of the above approaches individually
475 and compare the results in terms of accuracy. Then, we select the best method
to identify high-quality features.

4. User Identification: a Multi-Class Classification Application

In order to test the effectiveness and efficiency of the proposed feature eval-
uation and selection mechanism, we apply OFES in a multi-class classification
480 application, user identification based on sensory data. The goal of the applica-
tion is to identify a specific user from a group of n users based on their activity
pattern. This is a typical multi-class classification problem as each user expects
a unique label.

In the application, activity sensors including accelerometer and gyroscope,
485 are deployed on the user’s wrist. For each user, the readings of these activity
sensors are collected when the user walks normally. After applying basic data
preprocessing such as data normalization and cleaning, features are calculated
based on the collected data. OFES is then applied to evaluate and select a
small set of high-quality features. Based on the selected feature datasets, n -
490 user classification classifiers are trained. They are used to identify users when
needed.

5. Performance Evaluation

In this section, we present the performance evaluation of OFES when it is
applied in the above mentioned user identification application.

495 The performance of OFES is compared with two other popular feature evalu-
ation approaches, including Information Gain Feature Ranking (IGFR) + **Cor-**
relation Based Feature Subset Selection (CFSS) [4] and Random Pro-
jections with Matlab feature ranking [10]. The main ideas of these algorithms
are summarized in Section 2. We compare them in terms of the difference in

500 the selected feature sets, the impact of the features on accuracy of classifiers, the sensitivity to different classification algorithms, and the scalability of these algorithms in terms of the number of class labels.

In the rest of this section, we first describe the characteristics of the dataset and the definition of a list of 96 features that are evaluated using various different
 505 feature evaluation approaches. Then, we demonstrate the validity and necessity of feature evaluation and selection. **For all of our experiments, we set the window size of the sliding window algorithm as 500.** Finally, we illustrate the performance comparisons between OFES and others.

5.1. Description of Dataset and Feature Definition

510 Based on the MetaWear C sensing platform [45], we collected both accelerometer and gyroscope data from 30 users. During data collection, the Metawear device is attached to the wrist of the user, capturing the user’s hand movement while the user is walking around normally. Sensor readings consist of both accelerometer and gyroscope readings along x , y , and z axes consisting of
 515 6-tuples, $(A_x, A_y, A_z, G_x, G_y, G_z)$, where A_i and G_i specify the accelerometer and gyroscope on the i axis respectively. **For each class (i.e., each user), the data are collected in two sessions at two different times. Each session collects 15 seconds of data sampled at a frequency of 100Hz, corresponding to 1500 data points. Therefore, the whole dataset contains**
 520 **$30 \times 1500 \times 2 = 90000$ samples.** The raw data is then cleaned by denoising and normalization. **In addition, in our experiments, all participants are college students who may have similar activity patterns, which can be considered as a challenging case for multi-class classification.** Based on our literature search [4, 5, 10, 46, 47, 48, 49], we considered 17 different
 525 features (details are listed in Table 1) that are mostly used in biometrics identification for feature evaluation. Our activity sensor readings include both accelerometer and gyroscope readings along x , y , and z axes consisting of 6-tuples $(A_x, A_y, A_z, G_x, G_y, G_z)$. For each feature in Table 1, we extract 6 features along x , y , and z axes of both accelerometer and gyroscope readings. However,

530 features such as number of peaks is only for accelerometer and mean rotation rate is only for gyroscope readings. Thus, we extracted a total of 96 features, comprising 48 features from the accelerometer readings and 48 features from the gyroscope readings.

Table 1: The definition of features.

No.	Feature	Description
1.	Mean	Mean is a measure of central tendency calculated as the normalized sum of all the elements. $Mean(\mu) = \frac{\sum_{i=1}^n a_i}{n}$ Where, a = ax, ay, az, gx, gy, gz; n = the total number of data points
2.	Standard Deviation (Std)	Standard deviation is a measure that is used to quantify the amount of variation or dispersion of a set of data values. $Std(\sigma) = \sqrt{\frac{\sum_{i=1}^n (a_i - \mu)^2}{n}}$
3.	Root Mean Square (RMS)	RMS is defined as the square root of the mean square. $RMS = \sqrt{\frac{\sum_{i=1}^n (a_i)^2}{n}}$
4.	Mean Frequency (MNF)	The mean frequency of a spectrum is calculated as the sum of the product of the spectrogram intensity and the frequency, divided by the total sum of spectrogram intensity. $MNF = \frac{\sum_{i=1}^M f_i P_i}{\sum_{i=1}^M P_i}$
5.	Median	The median is the value separating the upper half of a sample data set from the lower half. In our case, n is even (i.e. n=1500) $Median = \frac{\left(\frac{n}{2}\right)^{th} \text{tern} + \left(\frac{n+1}{2}\right)^{th}}{2}$
6.	Median Frequency (MDF)	Median Frequency is a frequency that divides the power spectrum into two regions with equal amplitude. $MDF = \frac{1}{2} \sum_{i=1}^n P_i$
7.	Kurtosis	Kurtosis measures the tail-heaviness of the distribution. $kurtosis = \frac{1}{2} \sum_{i=1}^n \left[\frac{a_i - \mu}{\sigma} \right]^4$
8.	Skewness	Skewness is a measure of the symmetry in a distribution. $Skewness = \frac{1}{2} \sum_{i=1}^n \left[\frac{a_i - \mu}{\sigma} \right]^3$
9.	Peak to Peak	Peak to Peak computes the maximum minus minimum value in the time series. $PeaktoPeak = \max(a_i) - \min(a)_i$
10.	Peak to RMS	Peak to RMS results the ratio of maximum amplitude in the time series to RMS of the time series. $PeakToRMS = \frac{\max a_i }{RMS}$

No.	Feature	Description
11.	Variance	Variance is the average of the squared distances from each point to the mean. $\sigma^2 = \frac{\sum_{i=1}^n (a_i - \mu)^2}{n}$
12.	Average Peak Interval (API)	Let $I = (i_j, i_{j+1}, i_{j+2}, \dots, i_n)$ be a set of indices of peaks found in W_x . Then $API = \frac{1}{n-1} \sum_{j=1}^{n-1} (i_{j+1} - i_j)$
13.	Bandpower	The bandpower of a signal is defined as the average power in the given frequency range.
14.	Energy	The energy of a signal vector Wx is computed as $\int_j^i W \times (t)^2 dt$
15.	Number of Peaks (acceleration only)	Let $I = (i_j, i_{j+1}, i_{j+2}, \dots, i_n)$ be a set of indices of peaks found in Wx . Then the length(I) provides the number of peaks.
16.	Spectral Entropy	Spectral entropy describes the complexity of the signal. It is directly proportional to the peak of the signal power spectrum and similar to Shannon's entropy. With the use of the power spectral density as a probability density. It is computed as $\frac{1}{\log(N)} \sum p_i \log(p_i).$
17.	Mean Rotation Rate (rotation only)	Let $M = (\mu_1, \mu_2, \mu_3, \dots, \mu_r)$ be the vector containing the statistical mean of the absolute values between a consecutive pair of MS points in Wx . Then the Mean Rotation Rate (MRR) is defined as mean(M).

Based on the definition in Table 1, the feature dataset is calculated from the cleaned raw sensor dataset. For each specific feature, we will have two sessions of feature datasets. In the production of NRMSD matrix, the NRMSD value of diagonal entries in the matrix is calculated based on both sessions of data from the same user, while the NRMSD value of non-diagonal entries are calculated based on first session data of the user i and the second session data of the user j , where i, j are the index of the entry in the NRMSD matrix, and i is not equal to j .

When the dataset of selected features is used to construct the classifiers, the session one feature data is used as the training dataset and the session two feature data is used as the testing dataset.

5.2. Validity and Necessity of Feature Evaluation and Selection

Our goal of feature evaluation and selection is to identify a set of high-quality features, based on which we can construct classifiers with high accuracy, because we believe a small set of high-quality features are important and sufficient to build high-accurate classifiers.

To validate the above hypothesis, we conduct an experiment where a set of classifiers are constructed and evaluated using various classification algorithms and based on different number of selected features. We select four lightweight classification algorithms, which are widely used in user identification application, including K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Naive Bayes, and Random Forest Classifier. Selecting the total number of features is one of the important steps for classification since it impacts both accuracy and time complexity. Four different sets of features with sizes 8, 10, 12, and 14 are used to construct different classifiers. Accuracy of each classifier is evaluated.

Figure 3 shows the results of the above experiment, where x-axis specifies the classification algorithm, and y-axis indicates the accuracy of the classifier constructed using different number of selected features. Each colored line represents the accuracy of classifier constructed based on different classification algorithm for a specific number of selected features. From the figure, we observe that classifiers built based upon 4 different sets of features exhibit close performance in terms of accuracy, which is mostly between 90% and 96%. Classifiers built based on 8, 10, and 14 feature have very close performance for all four classification algorithms, while the classifier on 12 features has slightly less accuracy. Among the four classification algorithms, KNN exhibits best accuracy while Random Forest Classifier has the least accuracy. Some classification algorithm like Naive Bayes is less sensitive to the number of features while others such as Random Forest Classifier is more sensitive to it. The above observations confirms our belief that a small number of high-quality features is sufficient to build a highly accurate classifier. It is also necessary to identify a set of high-quality features to reduce the complexity of classification process.

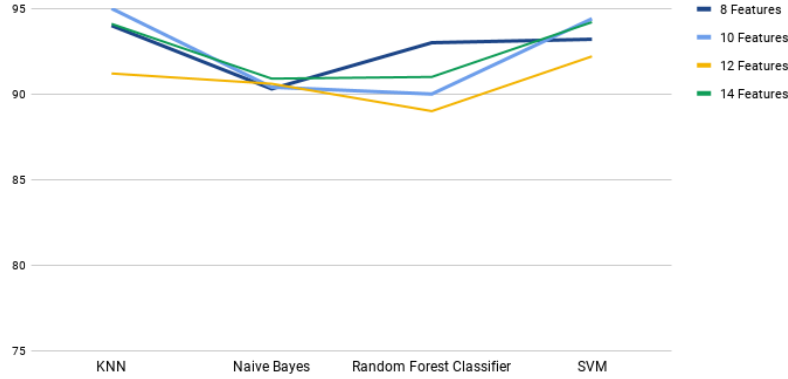


Figure 3: Impact of number of features on classification accuracy.

5.3. Performance comparison with other algorithms

ReliefF, **Sparsity Based Feature Selection**, Correlation Based Feature Selection (CFSS), Information Gain Feature Ranking (IGFR), and Random Projections are some of the common feature selection algorithms. The main ideas of these algorithms are summarized in Section 2. Among all the existing literature, [4, 5, 10] did an extensive job in providing comprehensive work on feature evaluation and selection, which are closely relevant to our research study. We compare our proposed feature evaluation and selection algorithm with these two best previous efforts by Kumar et al. [4] who used Information Gain Feature Ranking and Damaševičius et al. [10] who used Random Projections with Matlab feature ranking for feature selection. In addition, we also compare OFES with a feature selection approach that randomly selects a set of features from the feature set.

We select top ten ranked features in [4, 10] to compare with top 10 selected features of OFES. In addition, we also compare the performance of classifiers using features selected by above approaches and random selected features. For OFES, feature sets with both 8 feature and 10 feature are used. For the rest of the section, we use OFES(10) to refer OFES selecting 10 features and OFES(8) to refer OFES selecting 8 features.

595 In the rest of this section, we first compare the differences among top 10
selected featured using OFES, IGFR + **CFSS**, and Random Projections with
Matlab feature ranking. Then, for each compared feature selection approach,
we train classifiers based on four classification algorithms, including K-Nearest
Neighbors (KNN), Support Vector Machine (SVM), Naive Bayes, and Random
600 Forest Classifier using the set of features selected by that approach. The per-
formance of these classifiers reflects the performance of these feature evaluation
and selection approaches.

5.3.1. Feature evaluation results

Based on our feature evaluation method, we filter the features that have
605 $FV \geq 0.15$ and $FR \geq 15$ as high-quality features from the total of 96 extracted
features because the bigger FV and FR implies significant difference between
two classes.

From the filtered features based on specific values of FV and FR, we first
rank them from the highest to the lowest based on FV and FR separately. Then,
610 we rank them again considering both FV and FR, and select the top 10 and top
8 features out of 96 features.

5.3.2. Comparison of selected feature sets

The authors in [4] have extracted a total of 76 features (32 features from
the accelerometer readings and 44 features from the gyroscope readings). They
615 used information gain based feature ranking to rank the features. They ranked
the features of accelerometer and gyroscope separately [4], whereas OFES ranks
all the features at the same time.

The authors in [10] have extracted 99 features from the collected data based
on the extensive analysis of the literature and features used by other authors.
620 They used Random Projections for dimension reduction, and Matlab feature
ranking to rank the features.

Table 2 shows the comparison of top 10 features selected by different fea-
ture selection algorithms. In the table, ACC M and GYRO M are defined as

Table 2: Comparison of top 10 features selected by different feature selection algorithms.

Rank	OFES	IGFR + CFSS [4]	Random Projections + Matlab feature ranking [10]
1.	Mean ACC X	Median ACC Y	Variance GYRO Z
2.	Median ACC X	Energy ACC X	Variance ACC M
3.	Mean ACC Y	Energy ACC M	First eigenvalue of moving covariance of difference between ACC and GYRO $E_{ag} = eig_1(cov(a_x - g_x, a_y - g_y, a_z - g_z))$
4.	Median ACC Y	Median ACC Z	Energy GYRO Z
5.	Energy ACC X	Energy GYRO Z	Moving energy of difference between ACC Z and GYRO Z $ME_{ag} = \frac{1}{N} \sum_{i=1}^N (x_i - y_i)^2, here x = a_z, y = g_z$
6.	Median ACC Z	Median GYRO M	Variance ACC X
7.	Mean ACC Z	Mean Rotation Rate GYRO M	First eigenvalue of moving covariance between ACC $E_a = eig_1(cov(a_x(1:N), a_y(1:N), a_z(1:N)))$
8.	Energy ACC Y	Energy GYRO M	First eigenvalue of moving covariance between GYRO $E_g = eig_1(cov(g_x(1:N), g_y(1:N), g_z(1:N)))$
9.	Skew ACC Y	Mean Rotation Rate GYRO Z	Moving energy of orientation vector of ACC $MEA = \frac{1}{N} \sum_{i=1}^N \varphi_i^2, here \varphi = \frac{\arccos(a_x \cdot a_y)}{ a_x \cdot a_y }$
10.	RMS ACC X	Median ACC X	GYRO M

$\sqrt{(a_x)^2 + (a_y)^2 + (a_z)^2}$ and $\sqrt{(g_x)^2 + (g_y)^2 + (g_z)^2}$ respectively. We observe
 625 that there are no common features selected by all three algorithms. However
 there are four common features between OFES and IGFR + CFSS, including
 Median ACC X, Median ACC Y, Median ACC Z, and Energy ACC X, and
 none between OFES and Random Projections with Matlab Ranking. We also
 observe that OFES selected only accelerometer based features. Hence we can
 630 reduce the number of sensors by using only accelerometer sensor and improve
 cost efficiency of user classification based on activity sensor data.

5.3.3. Performance comparison in scalability to the number of class labels

In multi-class classification, the accuracy of classifier usually decreases with
 the increase in the number of class labels [18]. Classifiers constructed based on
 635 high-quality features should be scalable with the class labels. That is to say,
 the classifiers should maintain high accuracy with the increase of the number of

class labels.

In this experiment, we compare the performance of five feature selection approaches in 14, 20, 25, 30-class classification. We select four different classifiers including K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Naive Bayes, and Random Forest Classifier and select the best classifier with high accuracy. The results are depicted in Figure 4, where x-axis specifies the number of class labels, and y-axis indicates the accuracy of the best classifier constructed using the features selected by the corresponding feature selection algorithm. **The best classifiers for each method are described in Table 3.** Each colored line represents the accuracy of best classifier constructed based on the selected feature set for a specific number of class label.

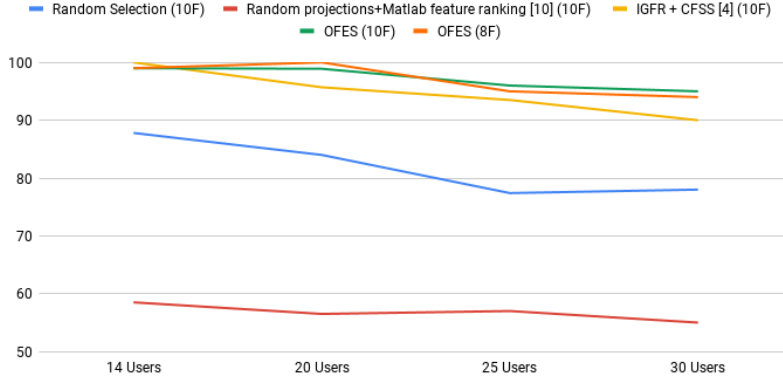


Figure 4: Performance comparison in scalability to the number of class labels

From the figure, we observe that accuracy of classification tends to decrease with increase in the number of class labels (users) for all algorithms. Among all approaches, IGFR + **CFSS** features results in the most significant accuracy drop, which is 10%, dropping from 100% for 14-user classification to 90% for 30-user classification. Following that is random selection features with a drop of 9.8%. Random Projections with Matlab feature ranking has similar scalability performance to OFES(10) and OFES(8). They all drop about 5% with the increase of number of users. However, the classifiers constructed using OFES(10)

and OFES(8) achieve way better accuracy than that using Random Projections with Matlab feature ranking.

5.3.4. Performance comparison in sensitivity to classification algorithms

Features selected with a feature selection algorithm should be less sensitive
 660 across different classifier models because each classifier will have its own advantages and disadvantages. Hence, there is a need for selected high-quality features to perform well with any kind of classifier model.

We select four classifier models, including K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Naive Bayes, and Random Forest Classifier in order
 665 to test the sensitivity of the feature selection algorithms. Figure 5 shows the comparison of sensitivity between different classifiers with respect to feature selection algorithms for 30-user classification. In the figure, the x-axis specifies different classification algorithms, and y-axis indicates the accuracy of the classifier constructed based on the corresponding classification algorithm by using the
 670 selected feature set. Each line in the figure denotes the accuracy of the classifier constructed by using feature set selected by a specific feature selection approach based on different classification algorithms. We compare the sensitivity based on standard deviation of the accuracy of classifier.

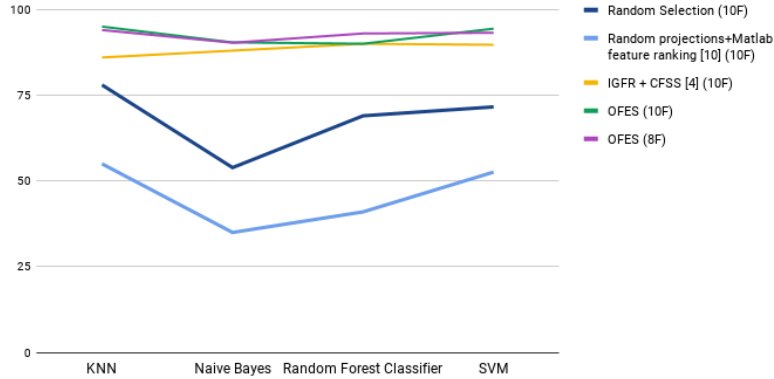


Figure 5: Performance comparison in sensitivity to classification algorithms.

From the figure, we observe that Random Projections with Matlab feature

675 ranking and random selection features exhibit high standard deviation of 8.22,
8.84 respectively while OFES (10), OFES (8) and IGFR + **CFSS** exhibit con-
sistent accuracy results with very small standard deviation of 2.26, 1.39, 1.59
respectively across different classifier models. The gap between accuracy of
the best and the worst classifiers for Random Projections with Matlab feature
680 ranking and random selection features ranges between 14 – 20% while for OFES
and IGFR + **CFSS**, it ranges between only 3 – 5%. This indicates OFES and
IGFR + **CFSS** are less sensitive to different classification algorithms. Com-
pared with IGFR + **CFSS**, OFES has higher accuracy in general. In summary,
OFES achieves both consistency and accuracy across different classifiers.

685 5.3.5. *Performance comparison in the impact of the features on accuracy of classifiers*

Selecting the high-quality features is necessary to accurately distinguish the
classes. We use accuracy of the classifier models as a measure to evaluate the
performance of feature selection algorithms. In order to compare the perfor-
690 mance of feature selection algorithms, for each feature selection algorithm, we
build four different classifiers including K-Nearest Neighbors (KNN), Support
Vector Machine (SVM), Naive Bayes, and Random Forest Classifier for 30 user
classification with the features selected by the corresponding feature selection
algorithm.

695 *Accuracy in 30-user classification.* In this section, we compare the overall per-
formance of feature selection algorithms for all the classifier models. Figure 6
shows the performance comparison in the impact of the features selected by
different feature selection algorithms on accuracy of classifiers in classifying 30
users. In the figure, the x-axis specifies different feature selection algorithms,
700 and y-axis indicates the accuracy of the classifier constructed by using the se-
lected feature set based on the corresponding feature selection algorithm. Each
line in the figure denotes the accuracy of the specific classifier constructed by
using feature set selected by different feature selection algorithms.

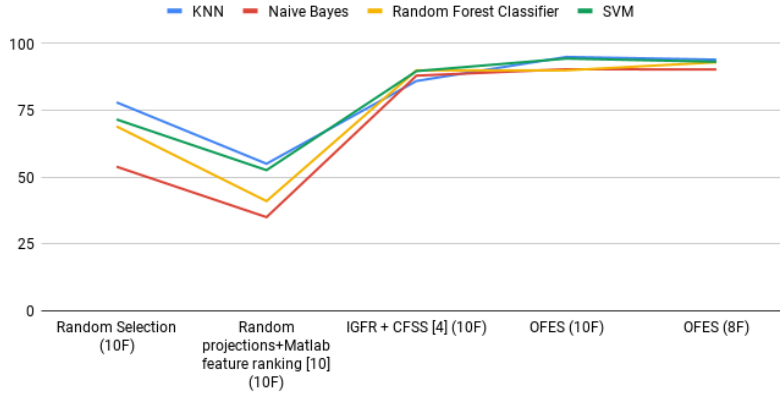


Figure 6: Performance comparison in the impact of the features on classification accuracy.

From the figure we observe that Random projections with Matlab ranking results in the least accurate results with an average accuracy of 45.9% among all classifier models. IGFR + **CFSS** with an average accuracy of 88.42% exhibits better results than random selection features and Random Projections with Matlab ranking, but it is not as good as OFES. OFES(10) achieves better results when compared to others for all the selected classifier models with an average accuracy of 92.45%. Even OFES(8) exhibits higher average accuracy of 92.62% when compared to others. We also observe that KNN is the best classification method for the majority of feature selection algorithms.

Best accuracy in 30-user classification. In this section, we compare the classifiers that achieve the highest accuracy for the corresponding feature selection algorithm. Table 3 shows the comparison of OFES with other approaches for 30-user classification in terms of accuracy and best classification method. With KNN as the best classification method, random selection features and Random Projections with Matlab ranking achieves accuracy of 78%, 55% respectively, while IGFR + **CFSS** achieves accuracy of 90% with Random Forest Classifier as the best classification method. OFES(8) and OFES(10) achieves higher accuracy of 94%, 95% respectively with KNN as the best classification method when compared to others.

From all above analysis, we conclude that OFES outperforms other feature evaluation and selection algorithms as it is less sensitive to different classification algorithms and more scalable to the number of class labels. It also achieves higher accuracy for most classification algorithms by identifying a smaller set of high-quality features.

Table 3: Performance comparison in accuracy of the best classification method.

Feature Selection Algorithm	Best Classification Method	Accuracy
Random selection	KNN	78%
Random Projections + Matlab feature ranking [10]	KNN	55%
Information Gain Feature Ranking (IGFR) +		
Correlation Based Feature Subset Selection (CFSS) [4]	Random Forest Classifier	90%
OFES(8)	KNN	94%
OFES(10)	KNN	95%

6. Conclusion and Future Work

In this paper, we proposed a novel and efficient approach, optimal feature evaluation and selection (OFES) for multi-class classification. Our proposed method discussed an approach for preliminary evaluation of different features before classifier model generation. We defined three measures: Normalized Root Mean Squared Difference (NRMSD), Farness Value (FV), and Farness Ratio (FR) to quantitatively measure the quality of features. Based on our feature evaluation and selection method, we discarded 90% of the total extracted features to improve the efficiency of the algorithm. As a result, a higher accuracy of 95% was achieved with the minimum number of features when compared to other approaches. Compared with other feature selection approaches, our approach also achieves better scalability to the number of class labels, and is less sensitive to the classification algorithms.

In future, we plan to extend OFES for integrated feature evaluation and selection as we observe the great potential when features are integrated from the NRMSD maps. We would also validate our design in applications with various types of features with different data types **and evaluate several factors,**
745 **such as a large number of user sets, spoofing, imbalanced datasets,**
etc. In addition, we would explore new mechanisms that combine
sparsity based feature selection with our approach, and perform sys-
tematic performance evaluation based on benchmark datasets. Finally,
we will design a new optimal multi-class classification algorithm by taking full
750 advantage of the optimal feature selection results.

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